

Vishal Kumar Allam

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PROFESSIONAL SUMMARY

- Quality Engineering professional with over **6+ years** of experience performing manual, exploratory, regression, API, integration, and automated testing across enterprise, healthcare, web, backend, and AI-enabled software applications.
- Strong programming and scripting experience with **Python, Java, C#, JavaScript/TypeScript, SQL, REST APIs, JSON, databases, Git, and automated testing frameworks**, supporting reliable software validation and defect investigation.
- Hands-on experience testing **AI and large language model (LLM) features**, evaluating outputs for accuracy, relevance, completeness, consistency, instruction-following, and expected behavior across varied prompts and contexts.
- Experienced creating AI-focused test scenarios, prompts, evaluation data, and regression checks to identify **hallucinations, incomplete responses, inconsistent outputs, context failures, and prompt-sensitive behavior**.
- Practical experience with **prompt engineering and context engineering**, including structured prompt variations, system/user context validation, boundary scenarios, and evaluation of how context influences AI-generated responses.
- Experienced using **Codex, Claude Code, GitHub Copilot, and similar AI-assisted development tools** to accelerate test creation, debugging, automation maintenance, defect investigation, and reusable QA workflow improvements.
- Strong analytical and communication skills with experience documenting defects using clear reproduction steps, expected and actual results, collaborating with developers and Product Owners, and supporting responsible AI and continuous QA improvement.

EDUCATION

University of Missouri Kansas City

"Master of Science" in Computer Science

Kansas City, MO

Jan. 2023 – May 2024

Jawaharlal Nehru Technological University

"Bachelor of Technology" in Computer Science

Hyderabad, India

Aug. 2015 - May 2019

TECHNICAL SKILLS

- **Programming & Scripting:** Python, Java, C#, JavaScript, TypeScript, SQL, JSON, REST APIs, Object-Oriented Programming, Bash, PowerShell.
- **Software Testing:** Manual Testing, Exploratory Testing, Regression Testing, Functional Testing, API Testing, Integration Testing, End-to-End Testing, Test Scenario Design, Test Data Preparation.
- **AI/LLM Testing:** Large Language Models, AI Product Evaluation, Accuracy, Relevance, Completeness, Consistency, Instruction-Following, Hallucination Detection, Incomplete Response Testing, AI Regression Validation.
- **Prompt & Context Engineering:** Prompt Engineering, Context Engineering, Prompt Variations, Context Validation, Boundary Testing, Negative Testing, Structured AI Test Scenarios, Evaluation Data Sets.
- **AI-Assisted Engineering:** Codex, Claude Code, GitHub Copilot, AI CLI Tools, AI-Assisted Test Authoring, Debugging, Test Maintenance, Defect Analysis, AI Evaluation Harnesses, Reusable Regression Checks.
- **API, Data & Tools:** Postman, Swagger/OpenAPI, REST/JSON, SoapUI, SQL Server, PostgreSQL, MySQL, Database Validation, Git, GitHub, Jira, Azure DevOps, Defect Tracking.
- **Quality Engineering Practices:** Agile/Scrum, SDLC, Defect Documentation, Root Cause Analysis, Test Documentation, Responsible AI, AI Risk Management, Ethical AI Awareness, Continuous Improvement, Cross-Functional Collaboration.

Test Automation Engineer

PCMI LLC (Full Time)

Oct 2024 – Present

Charlotte, NC, USA

- Execute manual, exploratory, regression, API, integration, and automated testing using **Python/C#, Playwright, Selenium WebDriver, REST APIs, JSON, and SQL**, ensuring reliable behavior across enterprise application workflows.
- Validate AI/LLM-enabled functionality by evaluating generated responses for **accuracy, relevance, completeness, consistency, and instruction-following**, using structured prompts, expected-response criteria, and repeatable evaluation scenarios.
- Create AI-focused test prompts, context variations, and test data to identify **hallucinations, incomplete responses, inconsistent results, context failures, and unexpected model behavior** across positive, negative, and boundary conditions.
- Apply prompt engineering and context engineering techniques to evaluate how instructions, supporting context, user inputs, and conversational history influence model responses and downstream application behavior.
- Use **Codex, Claude Code, and AI-assisted engineering tools** to accelerate Playwright test creation, debug failed executions, generate reusable test utilities, analyze logs, and improve regression-test maintainability.
- Support AI evaluation harnesses and reusable regression checks by automating prompt execution, response capture, deterministic validations, API assertions, and comparison of AI outputs against defined acceptance criteria.
- Document defects in Jira/Azure DevOps with clear reproduction steps, expected versus actual results, logs, screenshots, and technical evidence while collaborating with developers, Product Owners, and QA engineers through Agile delivery.

Environments: Python, C#, Playwright, Selenium WebDriver, REST APIs, JSON, Postman, SQL Server, LLM Testing, Prompt Engineering, Context Engineering, Codex, Claude Code, AI Evaluation Harnesses, Jira, Azure DevOps, Git, Agile/Scrum, Responsible AI.

SDET / QA Automation Engineer

Client: CVS

Dec 2023 – Sept 2024

Charlotte, NC, USA

- Performed manual, exploratory, regression, API, integration, and automated testing using **Java, Python, TypeScript, Playwright, Selenium WebDriver, RESTAssured, JSON, and SQL** across healthcare applications and backend services.
- Tested AI-assisted application capabilities by reviewing generated outputs for accuracy, relevance, completeness, consistency, and compliance with expected instructions and healthcare business rules.
- Designed prompts, test data, negative scenarios, and context variations to expose hallucinations, incomplete responses, inconsistent AI results, edge cases, and failures triggered by ambiguous or insufficient context.
- Performed API and database validation using **Postman, RESTAssured, JSON payloads, SQL, and backend reconciliation**, ensuring data consistency and correct interaction between application, service, and AI-enabled components.
- Used AI-assisted coding tools including Copilot/Claude-style workflows to generate test ideas, accelerate automation development, debug failures, improve reusable components, and identify additional regression coverage.
- Integrated automated regression checks into Jenkins/Git-based CI/CD workflows and investigated failed tests through application logs, API responses, SQL queries, Splunk, and structured root-cause analysis.
- Created detailed defect reports with reproduction steps, expected and actual results, evidence, and business impact while collaborating with developers, QA engineers, analysts, and Product stakeholders in Agile/Scrum teams.

Environments: Java, Python, TypeScript, Playwright, Selenium WebDriver, RESTAssured, Postman, JSON, SQL, Jenkins, Git, Splunk, Jira, AI/LLM Testing, Prompt Validation, AI-Assisted Testing, API Testing, Regression Testing, Agile/Scrum.

Software Engineer

Infor

Sept 2021 – Dec 2022

Hyderabad, India

- Designed and executed manual, exploratory, functional, regression, API, and integration testing using **Java, Python, Selenium WebDriver, Cucumber BDD, JUnit, TestNG, REST APIs, and SQL** across enterprise applications.
- Translated user stories, functional specifications, and acceptance criteria into reusable test scenarios, test data, positive/negative cases, and regression coverage while identifying gaps and potential quality risks.
- Performed REST/SOAP API and backend validation using **Postman, SoapUI, JSON, XML, and SQL**, verifying request/response behavior, data transformations, business rules, and integration consistency.

- Developed Python and Java utilities to automate repetitive test activities, validate structured application data, analyze responses, and improve maintainability of integration and regression-testing workflows.
- Investigated defects using application logs, API payloads, SQL data, execution evidence, and debugging techniques while documenting clear reproduction steps, expected results, actual results, and regression impact.
- Integrated automated tests with **Jenkins, Git, Maven, and CI/CD pipelines**, supporting repeatable regression execution, automated build validation, and faster feedback throughout Agile development cycles.
- Collaborated with developers, QA engineers, analysts, and Product stakeholders during Scrum ceremonies, defect triage, test reviews, and release activities while contributing reusable test documentation and process improvements.

Environments: Java, Python, Selenium WebDriver, Cucumber BDD, JUnit, TestNG, Postman, SoapUI, REST/SOAP APIs, JSON/XML, SQL, Jenkins, Git, Maven, Jira, CI/CD, Manual/Exploratory Testing, Defect Management, Agile/Scrum.

Associate Engineer QA

June 2019 – August 2021

Virtusa

Hyderabad, India

- Executed manual and automated functional, exploratory, regression, API, and integration testing using **Java, Selenium WebDriver, Cucumber BDD, JUnit, TestNG, REST APIs, and SQL** across enterprise web and backend applications.
- Created reusable test scenarios and test data from user stories and acceptance criteria, covering positive, negative, boundary, integration, and regression conditions throughout the software development lifecycle.
- Validated APIs and backend services using **Postman, SoapUI, REST/SOAP interfaces, JSON/XML payloads, and SQL queries**, ensuring accurate responses, data integrity, error handling, and downstream processing.
- Investigated failed tests and application defects using logs, screenshots, API responses, database queries, and debugging evidence while distinguishing automation issues from genuine product defects.
- Documented defects in Jira with clear steps to reproduce, expected behavior, actual behavior, severity, and supporting evidence while coordinating with developers to verify fixes and complete targeted regression testing.
- Integrated automated regression suites with **Jenkins, Git, and CI/CD pipelines**, supporting consistent build validation, repeatable test execution, and earlier identification of software quality issues.
- Participated in Agile/Scrum ceremonies and collaborated with developers, analysts, and QA team members to improve test documentation, reusable test cases, defect-prevention practices, and overall QA process effectiveness.

Environments: Java, Selenium WebDriver, Cucumber BDD, JUnit, TestNG, Postman, SoapUI, REST APIs, JSON/XML, SQL, Jenkins, Git, Jira, CI/CD, Manual Testing, Exploratory Testing, Regression Testing, API/Integration Testing, Agile/Scrum.

AWARDS & ACHIEVEMENTS

- 2019-11 Oracle Certified Associate, JAVA SE8 Programmer.
- ISTQB Certified Professional.
- Microsoft Technology Associate (MTA) – Database Administration Fundamentals, HTML- 5 Application Development Fundamentals.
- Awarded “Software Process Achievement” Award for my contribution towards the project in Virtusa.
- Received Customer Appreciation mails for quality releases.